

H50

**Variable Speed Pan Tilt
User Manual**

Thank you for using our products, please read the user manual carefully before using this product, this manual will provide you with the correct product parameters, instructions and precautions!

If you encounter any problems not mentioned in this manual, please contact us and we will provide you with satisfactory technical support.

Our company reserves the right to modify and update the manual due to product performance improvement, and the updated content will not be notified to users.

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1.Important Safeguards and Warnings

1.1 Warning:To remind users of the potential for serious injury!

- 1.1.1 Before installing the device, read the user manual carefully and keep it safe for future use.
- 1.1.2 All warning items on equipment and instructions should be followed.
- 1.1.3 Please use the type and voltage of power supply specified in the manual.
- 1.1.4 Ensure that the main power switch of the device is turned off before connecting or disconnecting the input and output cables to avoid personal injury and device damage.
- 1.1.5 Protect the input and output power cables to avoid personnel injury and equipment damage caused by leakage or short circuit.
- 1.1.6 To prevent lightning strikes, install standard lightning protection devices.
- 1.1.7 The equipment should be installed on a firm platform or bracket to avoid personal injury and equipment damage.
- 1.1.8 Ensure that there are no obstacles around the device to prevent device damage caused by blocking the device rotation path.
- 1.1.9 Please do not install or use the head in places that are too humid, dusty or smoky or near heat sources.
- 1.1.10 If the head gives out abnormal smell or smoke, stop using immediately, and cut off the power supply, and then contact our company.
- 1.1.11 Opening the case without authorization may cause electric shock or damage to the equipment.
If you have any problems, please do not try to repair by yourself. Please contact us for all maintenance matters. Do not use organic solvent to wipe the machine shell, to avoid affecting the sealing of the machine and resulting in water.

1.2 Caution

- 1.2.1 During transportation, storage, and installation, prevent heavy pressure, severe vibration, and water immersion; otherwise, the equipment may be damaged.
- 1.2.2 The head should be kept away from vibration or magnetic interference.
- 1.2.3 Waterproof treatment should be paid attention to when the top wires of the PTU are connected with the customer loading equipment, so as to avoid rainwater from the inside of the PVC pipe into the PTU.
- 1.2.4 Rs-485 and video signals should keep enough distance from high-voltage equipment or cables during transmission, and take protective measures against lightning strike and surge when necessary.

1.2.5 In order to prevent lightning strikes, please install a lightning arrester. The lightning protection GND of the PTU must be reliably grounded in accordance with the NATIONAL standard GB50198-94 (civil closed-circuit television system engineering technical specifications), If lightning protection and grounding measures are not taken, the PTU may be damaged. The resistance between the ground point and the PTU GND must be less than 4Ω .

1.2.6 Please use the power supply that meets the requirements of the head, otherwise the device will be damaged. 1.2.7 After the head is powered off, please restart it at an interval of 15 seconds.

Note: insulation treatment must be carried out on all external leakage wiring harness of pylon before power on, in order to avoid the damage of products caused by short circuit after power on!

2 Product introduction

2. Product Overview

2.1 Introduce design

As a high-end security surveillance product, Outdoor High-Duty Variable Speed Pan/Tilt has construction of high-strength aluminum alloy frame and modified plastic cover. It features anti-wind design, power off protection (self-lock) and smooth motion. It also features variable speed, 360 degrees continuous movement, auto scan, auto cruise, keep watch and OSD menu...etc. This Pan/Tilt is specially used for surveillance in far-field, like application in field of fire-fighting in forest, coast and border, bridges over sea or river and freeway.etc

2.2 Application fields

2.2.1 Forest fire prevention

The PTU is designed with wind resistance, suitable for wild strong wind and harsh environment. The waterproof grade enables the holder to be used in rain, snow, fog and high humidity environment. The shell is made of high-strength aluminum alloy, which can withstand certain physical impact. Heavy load, can carry high multiple lenses and other monitoring equipment.

2.2.2 Highway surveillance

The PTU has stable performance and can operate normally in the high and low temperature environment of $-25^{\circ}\text{C} \sim +65^{\circ}\text{C}$. The PTU has reliable variable speed and is suitable for expressway monitoring occasions.

2.2.3 Application of bridge across river (sea)

The PTU is waterproof, salt spray resistant(optional), corrosion resistant and other design, and

the pitching Angle range is large, horizontal 360 degrees unlimited position, observation range is large, suitable for river, sea and other monitoring.

2.2.4 Application for the car (boat)

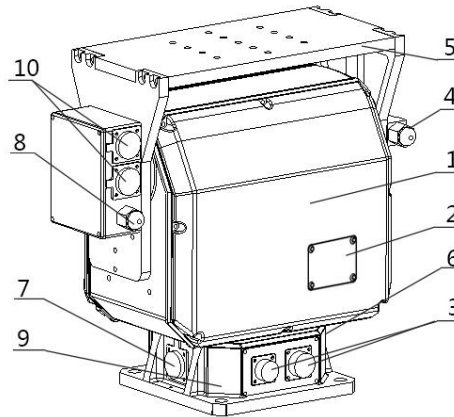
The body and shell of the PTU are made of high-strength aluminum alloy, with anti-seismic design and self-locking function during power failure, making the PTU suitable for vehicle applications.

2.2.5 Special application field

The design of the PTU can meet special customization requirements, no matter the structure circuit or function, can be customized for special application occasions, suitable for laboratory, research institute, teaching demonstration, auxiliary development, etc.

3 Installation instructions

3.1 Structure declaration



- | | | |
|--------------------|---------------------------------------|-------------------------|
| 1. Main body | 2.Back door for switch configuration. | 3.Bottom aviation plugs |
| 4.Top wires(right) | 5.Mounting plate | 6.Bottom USB3.0(right) |
| 8.Top wires(left) | 9.base | 7.Bottom USB3.0(left) |
| | 10.Top USB3.0 | |

Note: Only used for this customized H50 structure description

3.2 Confirmation before installation

Before installing the product, confirm the installation size of the PTU base and support. This product can be rotated continuously horizontally at 360°. When installing camera housing or other

top load products, ensure that the installation site has enough space for this product and its mounting components.

Bearing condition confirmation, because the product contains rotating parts, it must be confirmed that the product installation has sufficient strength, to ensure safety, the support supporting the head and cover should bear at least 2 times the total weight of the PTU and loading.

External protection device confirmation, according to the NATIONAL standard GB50198-2011 (civil closed-circuit television system engineering technical specifications), when the voltage fluctuation exceeds the range of $-15\% \sim +10\%$, should be set up the regulator power supply device, the regulator device nominal power shall not be less than 1.5 times the use of power; Lightning protection devices should be installed for the indoor ends of overhead cables entering the monitoring room and cable ends of products installed in the wilderness, tower tops or above nearby buildings.

Confirm the PTU power supply cable. Since the product has relatively high power and low voltage transmission power supply, if the distance between the PTU and the end output of the power supply adapter is far, you must consider the power loss of the cable. For specific requirements of the power supply cable, refer to "AC24V Cable Diameter and Transmission Distance Relation Table".

3.3 Installation procedure

3.3.1 Before installation, please check the accessories listed in the packing list and read the user manual carefully. If any accessories are missing or the description is not clear, please contact us.

3.3.2 Fix the head holder to a solid platform or base, using the screws in accessories and tighten the four screws of the base.

3.3.3 Remove the mounting plate, fix the top-load equipment on the mounting plate, and then fix the equipment on the PTU through the mounting plate.

3.3.4 Connect the top wires to the top-load equipment.

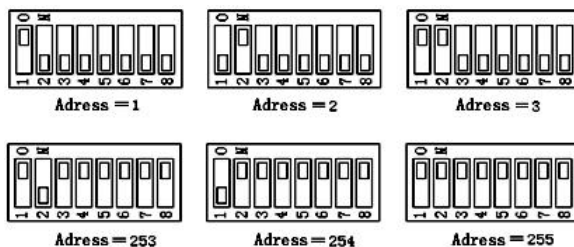
3.3.5 Connect the aviation plug to the power supply, back end monitor or computer, and connect the control wires to the keyboard or computer.

3.3.6 After the above steps are completed, the power can be used.

3.4 Parameter setting

When setting the address and function dip switches, open the rear cover of the PTU

3.4.1 Address setting



Address	The state of SW1							
	DIP-1	DIP-2	DIP-3	DIP-4	DIP-5	DIP-6	DIP-7	DIP-8
1	ON	OFF	OFF	OFF	OFF	OFF	OFF	OFF
2	OFF	ON	OFF	OFF	OFF	OFF	OFF	OFF
3	ON	ON	OFF	OFF	OFF	OFF	OFF	OFF
4	OFF	OFF	ON	OFF	OFF	OFF	OFF	OFF
5	ON	OFF	ON	OFF	OFF	OFF	OFF	OFF
...	...							
254	OFF	ON	ON	ON	ON	ON	ON	ON
255	ON	ON	ON	ON	ON	ON	ON	ON

Only part of the address codes are listed above, and the rest follow the binary code analogy.

3.4.2 Baud rate setting

The PUT provides various communication baud rates such as 2400bps, 4800bps, 9600bps and 19200bps, You can set the first and second bits of SW2 as follows:

Baud rate	The state of SW2	
	SW2-1	SW2-2
2400	OFF	OFF
4800	ON	OFF
9600	OFF	ON
19200	ON	ON

3.4.3 Setting communication Protocols

The third bit of SW2 is used to select the control protocol of the PTU. The setting method is as follows:

Protocols	SW2-3
-----------	-------

Pelco-D	OFF
Pelco-P	ON

Note: PELCO-D and PELCO-P are general control protocols, which can be controlled by control keyboard and general control software. The control device protocol must match the PTZ setting protocol.

3.4.4 Cruise preposition lens match

The fourth position of SW2 of the internal circuit board is to set the cruising state of the PTU. When the PTU is used normally with the lens feedback, please set this position to OFF. If the PTU is not connected to the lens feedback and the cruise function is tested separately, please set this position to ON.as follows:

Use condition	SW2-4
With lens feedback wires	OFF
Without lens feedback wires	ON

3.4.5 Control interface Settings (422 control interface only)

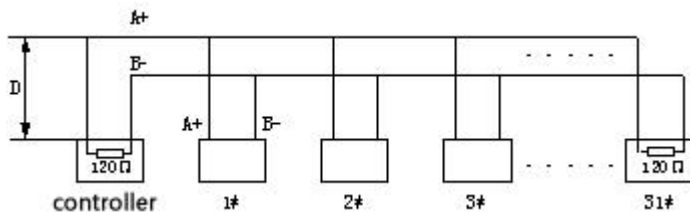
When the PTU needs to be set as RS422 control interface, the 5th position of SW2 should be switched to ON, and the 1st and 2nd positions of SW3 should be switched to OFF at the same time. When the PTU needs to be set as the RS485 control port, switch the 5th position of SW2 to OFF, and switch the 1st and 2nd positions of SW3 to ON at the same time. The setting method is as follows:

Control interface	State of SW2	State of SW3	
	SW2-5	SW3-1	SW3-2
RS485	OFF	ON	ON
RS422	ON	OFF	OFF

Note:Please install the back cover after the address code, protocol code and baud rate are selected, and pay attention to the silicone sealing strip at the cover plate not to be lost or missing.

3.4.6 Connection mode and terminal resistance

a)RS485 industrial bus standard requires all devices to adopt chrysanthemum chain connection mode, both ends must be connected with 120 Ω terminal resistance, and the distance between the "D" segment shall not exceed 7 meters, as shown below:



b)The connection mode of 120Ω resistor of the device terminal is as follows:The control circuit board is equipped with equipment terminal resistance 120Ω resistance , There are two types of connection.The connection mode is set to the default before delivery , At this point, put third position of SW3 on circuit board OFF (down) state.The 120Ω resistor is not connected.When a 120Ω resistor needs to be connected,put third position of SW3 on circuit board ON (UP) state,So the 120Ω resistor is plugged into the circuit.

Note: if the cloud platform is RS422 control interface, the 4th bit of SW3 must be operated together with the 3rd bit of SW3.

4 Enhanced function of the PTU

Note:This series of functions can be controlled by display keyboard or interface software,It can also be controlled by sending instructions directly;for example PelcoD command,the corresponding command format is as follows:(**Note: XX is in hexadecimal format**)

Control command format:

Bit1	Bit 2	Bit 3	Bit 4	Bit 5	Bit 6	Bit 7
The starting byte	Address	Command byte 1	Command byte 2	data1	data2	The ending byte
FF	add	command1	command2	data1	data2	check code

Command 1 and 2 are as follows:

	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
Command 1	0	0	0	0	0	Iris Close	Iris Open	Focus Near
Command 2	Focus Far	Zoom Wide	Zoom Tele	Down	Up	Left	Right	0

check code = [(bit2 + bit3 + bit4 + bit5 + bit6) & 0xFF]

Data codes data1 and data2 represent horizontal and vertical speed respectively (00-40H)

Example: Run at maximum speed 0x40, address 0x01 command as following:

{0xFF,0x01,0x00,0x02,0x40,0x00,0x43,} //right

{0xFF,0x01,0x00,0x04,0x40,0x00,0x45,} //lift

{0xFF,0x01,0x00,0x08,0x00,0x40,0x49,} //up

{0xFF,0x01,0x00,0x10,0x00,0x40,0x51,} //down

Note: The corresponding stop orders above are {0xFF,0x01,0x00,0x00,0x00,0x00,0x01}

When the PTZ is stationary, the serial port is receiving and does not send back data.

Turn on auxiliary switch: FF Add 00 09 00 XX Checksum

Turn off auxiliary switch: FF Add 00 0B 00 XX Checksum

Set preset: FF Add 00 03 00 XX Checksum

Delete preset: FF Add 00 05 00 XX Checksum

Call preset: FF Add 00 07 00 XX Checksum

● Turn on/off auxiliary switch

- 1.Turn on/off auxiliary switch No.1 Off by default
- 2.Turn on/off auxiliary switch No.2 Off by default
- 3.Turn on/off auxiliary switch No.3 Customized according to customer requirements
- 4.Turn on/off auxiliary switch No.4 Customized according to customer requirements
- 5.Turn on/off auxiliary switch No.6 Turn on/off Angle real-time return (limited to RS422 control interface)

● Cruise time setting

- 1.Set No.91 preset position Set the stopping time between each point of cruise as 4s
- 2.Set No.92 preset position Set the stopping time between each point of cruise as 10S(default)
- 3.Set No.93 preset position Set the stopping time between each point of cruise as 20s
- 4.Set No.94 preset position Set the stopping time between each point of cruise as 30s
- 5.Set No.95 preset position Set the stopping time between each point of cruise as 1min
- 6.Set No.96 preset position Set the stopping time between each point of cruise as 2min
- 7.Set No.97 preset position Set the stopping time between each point of cruise as 3min
- 8.Set No.98 preset position Set the stopping time between each point of cruise as 4min

● Call cruise

- 1.Call No.91 preset position Turn on the first cruise route,call No.1-No.10 preset

position

.....

2.call No.98 preset position Turn on the eighth cruise route,call No.71-No.80 preset position

● **PTU reference point setting**

Set No.0(No.99) preset position,After setting successfully , The reference point will be the position of the preposition NO.0 (No.99), That is, the reference point for the Angle return and Angle positioning of the PTU, This feature does not affect the preset position previously set, The PTZ will automatically return to 0 (99) preset position after self-test;

● **Remote restart function**

Call the No.100 preset position Turn on the remote restart function, the PTU will self-test

● **Power-on self-check function (It can be customized according to customer requirements)**

1.Set No.101 preset position Turn on self-test

2.Delete No.101 preset position Turn off self-test

3.Call No.101 preset position Perform a re-test

● **Set parameters to restore default values**

Set No.102 preset position Set parameters to restore default values

● **Auto scan setup and call**

1.Set No.103 preset position Set the start point of auto scan

2.Set No.104 preset position Set the end point of auto scan

3.Call No.103 preset position Turn on auto scan

4.Call No.104 preset position Turn off auto scan

● **Speed setting for preset position and auto scan**

1.Set No.105 preset position Preset position speed increases

2.Delete No.103 preset position Preset position speed decreases

3.Set No.106 preset position Auto scan speed increases

4.Delete No.106 preset position Auto scan speed decreases

● **Keep Watch function**

1.Set No.107 preset position Turn on keep watch,the PTU will automatically call preset No.1 position at the set time after all actions are completed

2.Delete No.107 preset position Turn off keep watch

- **Keep Watch time setting**

- | | |
|------------------------------|--|
| 1.Set No.108 preset position | Set the automatic homing time to 1min |
| 2.Set No.109 preset position | Set the automatic homing time to 2min (default) |
| 3.Set No.110 preset position | Set the automatic homing time to 5min |
| 4.Set No.111 preset position | Set the automatic homing time to10min |
| 5.Set No.112 preset position | Set the automatic homing time to20min |
| 6.Set No.113 preset position | Set the automatic homing time to30min |
| 7.Set No.114preset position | Set the automatic homing time to45min |
| 8.Set No.115 preset position | Set the automatic homing time to60min |

- **Real-time return function (Under RS232 or RS422)**

- | | |
|-------------------------------|---------------------------------|
| 1.Call No.116 preset position | turn on real-time angle return |
| 2.Set No.116 preset position | turn off real-time angle return |

- **Clearance compensation function**

- | | |
|---------------------------------|---|
| 1.Set No.117 preset position | Restore the compensation value of horizontal and vertical clearance |
| 2.Set No.118 preset position | Increase the horizontal clearance compensation value |
| 3.Delete No.118 preset position | Reduce the horizontal clearance compensation value |
| 4.Set No.119 preset position | Increase the vertical clearance compensation value |
| 5.Delete No.19 preset position | Reduce the vertical clearance compensation value |

- **Step by step scanning Settings and calls(Customize according to customer requirements)**

- | | |
|---------------------------------|------------------------------|
| 1.Set No.120 preset position | Set auto scan starting point |
| 2.Delete No.120 preset position | Set auto scan ending point |
| 3.Call No.120 preset position | Start auto scan |

- **Keep watch (Customize according to customer requirements)**

- | | |
|---------------------------------|---|
| 1.Set No.122 preset position | Set keep watch mode to call No.1 position |
| 2.Delete No.122 preset position | Set keep watch mode to call No.1 cruise |
| 3.Call No.122 preset position | Set keep watch mode to call No.1 scanning |

5 PTU control Command

No.	The name of the function	Command	Remark
1	Horizontal Angle query command	FF Addr 00 51 00 00 Checksum	
2	Vertical Angle query command	FF Addr 00 53 00 00 Checksum	
3	Horizontal Angle return format	FF Addr 00 59 DataH DataL Checksum	①
4	Vertical Angle return format	FF Addr 00 5B DataH DataL Checksum	②
5	Horizontal Angle positioning	FF Addr 00 4B DataH DataL Checksum	①
6	Vertical Angle positioning	FF Addr 00 4D DataH DataL Checksum	②
7	positioning speed setting	FF Addr 00 5F HSPD VSPD Checksum	③
8	Set auto scan starting point	FF Addr 00 11 00 00 Checksum	
9	Set auto scan ending point	FF Addr 00 13 00 00 Checksum	
10	Run horizontal auto scan	FF Addr 00 1B 00 00 Checksum	
11	Stop horizontal auto scan	FF Addr 00 1D 00 00 Checksum	
12	Scanning speed setting	FF Addr 00 31 HSPD VSPD Checksum	③
13	Vertical step setting	FF Addr 00 33 DataH DataL Checksum	④
14	Cruise residence time setting	FF Addr 00 69 00 time Checksum	⑤
15	Cruising speed setting	FF Addr 00 67 HSPD VSPD Checksum	
16	Running cruise function	FF Addr 00 4F 00 data Checksum	⑥
17	Zoom value query	FF Addr 00 55 00 00 Checksum	
18	Zoom value return	FF Addr 00 5D DataH DataL Checksum	
19	Focus value query	FF Addr 00 65 00 00 Checksum	
20	Focus value return	FF Addr 00 6D DataH DataL Checksum	
21	Zoom value positioning	FF Addr 00 4F DataH DataL Checksum	Value greater than 7
22	Focus value positioning	FF Addr 00 6F DataH DataL Checksum	
23	Set homing time	FF Addr 00 6B 00 time Checksum	⑦
24	Restore factory default parameters	FF Addr 00 29 00 00 Checksum	
25	Remote restart	FF Addr 00 0F 00 00 Checksum	
26	Preset position peed query	FF Addr 00 71 00 00 Checksum	
27	Preset position peed return	FF Addr 00 71 HSPD VSPD Checksum	
28	Positioning speed query	FF Addr 00 73 00 00 Checksum	

29	Positioning speed return	FF Addr 00 73 HSPD VSPD Checksum	
30	auto scan speed query	FF Addr 00 75 00 00 Checksum	
31	auto scan speed return	FF Addr 00 75 HSPD VSPD Checksum	
32	Cruise stay time query	FF Addr 00 77 00 00 Checksum	
33	Cruise stay time return	FF Addr 00 77 00 time Checksum	
34	Keep Watch time query	FF Addr 00 79 00 00 Checksum	
35	Keep Watch time return	FF Addr 00 79 00 time Checksum	
36	Horizontal error compensation query	FF Addr 00 7B 00 00 Checksum	
37	Horizontal error compensation return	FF Addr 00 7B DataH DataL Checksum	
38	Vertical error compensation query	FF Addr 00 7D 00 00 Checksum	
39	Vertical error compensation return	FF Addr 00 7D DataH DataL Checksum	

6 Troubleshooting

The following is a list of users in the process of using the head may encounter some problems, you can refer to the table to solve their own or contact our company, we will provide you with satisfactory technical support and services.

The fault phenomenon	Possible reasons	The solution
no self-check after power-on	Power adapter damaged	Replace the power supply
	Wrong connection of power supply	Reconnect the power cable.
	Engineering wires failure	Check and troubleshoot
The self-test is abnormal	The power of the power adapter is insufficient	Replace the power supply
	Mechanical failure	Need to repair
The video image is unstable	Poor wire contact	rewiring
After power-on, the self-test is successful, but the head can not be controlled	The control line is connected inversely	rewiring
	The address number, protocol, or baud rate of the PTU is incorrect	Reset the PTU dip switch
	The keyboard address, protocol, or baud rate Settings do not match the head	Reset the keyboard or control software

7 Quality assurance and after-sales service

- 7.1 Customer satisfaction is our purpose, quality is the life of our company, our PTU products are using a number of military technology and unique key technology, advanced performance, reliable quality.
- 7.2 As long as you request, we can rely on advanced technology, rich experience and customer-centric service concept for you to customize a set of quality, practical system.
- 7.3 Our products are equipped with detailed installation instructions and operation instructions, installation and operation is convenient, and in the case of conditions allow, our company's technical staff can provide technical support.
- 7.4 In the process of using our products, if you encounter any quality, technology, operation and other problems, welcome to contact our company, we will solve the problem for you in the shortest time.
- 7.5 Welcome customers and friends to give us all aspects of the comments and suggestions, your support is always the power of our development, thank you!

Appendix I AC24V Table of relation between wire diameter and transmission distance

When the wire diameter is constant and the AC24V voltage loss rate is less than 10%, the recommended maximum transmission distance. For ac power supply equipment, the maximum allowable voltage loss rate is 10%. For example, if the rated power of a device is 80W and the device is installed 35 feet (10m) away from the transformer, the minimum wire diameter is 0.8000mm.

Transmission Distance (feet) Wire diameter (mm) Power(w)	0.8000	1.000	1.250	2.000
10	283 (86)	451 (137)	716 (218)	1811 (551)
20	141 (42)	225 (68)	358 (109)	905 (275)
30	94 (28)	150 (45)	238 (72)	603 (183)
40	70 (21)	112 (34)	179 (54)	452 (137)
50	56 (17)	90 (27)	143 (43)	362 (110)
60	47 (14)	75 (22)	119 (36)	301 (91)
70	40 (12)	64 (19)	102 (31)	258 (78)
80	35 (10)	56 (17)	89 (27)	226 (68)
90	31 (9)	50 (15)	79 (24)	201 (61)
100	28 (8)	45 (13)	71 (21)	181 (55)
110	25 (7)	41 (12)	65 (19)	164 (49)
120	23 (7)	37 (11)	59 (17)	150 (45)
130	21 (6)	34 (10)	55 (16)	139 (42)
140	20 (6)	32 (9)	51 (15)	129 (39)
150	18 (5)	30 (9)	47 (14)	120 (36)
160	17 (5)	28 (8)	44 (13)	113 (34)
170	16 (4)	26 (7)	42 (12)	106 (32)
180	15 (4)	25 (7)	39 (11)	100 (30)
190	14 (4)	23 (7)	37 (11)	95 (28)
200	14 (4)	22 (6)	35 (10)	90 (27)

Appendix II Comparison table of domestic and foreign wire gauge

Metric bare wire diameter (mm)	Approximate American wire gauge AWG	Approximate English wire gauge SWG	Cross sectional area of bare wire (mm ²)
0.100	38	42	0.00785
0.110	37	41	0.00950
0.130	36	39	0.01327
0.140	35		0.01539
0.160	34	37	0.02011
0.180	33		0.02545
0.200	32	35	0.03142
0.230	31		0.04115
0.250	30	33	0.04909
0.290	29	31	0.06605
0.330	28	30	0.08553
0.350	27	29	0.09621
0.400	26	28	0.1237
0.450	25		0.1602
0.560	24	24	0.2463
0.600	23	23	0.2827
0.710	22	22	0.3958
0.750	21		0.4417
0.800	20	21	0.5027
0.900	19	20	0.6362
1.000	18	19	0.7854
1.250	16	18	1.2266
1.500	15		1.7665

Appendix III Technical specification

Model	H50
Speed	Pan :0.01° ~ 30°/s ; Tilt:0.01° ~ 15°/s
Rotation Range	Pan:±180° ; tilt:+90° ~ -90°
Preset	200 个
Presetting Precision	±0.1°
Auto Cruise	8 tracks(each with 10 presets)
Homing time	It can be set up within the scope of 1~60 min
Protocol	PELCO_D/P
Baud rate	9600bps (2400bps/4800bps optional)
Communication	RS485
Input Voltage	DC24V
Power Consumption	≤80W
Operation Temperature	-40°C~+65°C,90%±3%(without condensation)
Storage Temperature	-45°C ~ +65°C
Max. Load	50KG
Dimensions (L×W×H)	384mm×185mm×361mm
Weight	22.5kg
Load mode	Top load
Driven way	Worm-gear
Waterproof	IP66

Appendix IV cables definition

Bottom cables definition		
7 core aviation plug(left)		
No.	Color	Definition
1	Brown 1.0AF wire	DC24V+
2	Blue 1.0AF wire	GND
3	Yellow AF wire	RS485-B
4	Black 1.0AF wire	RS422-GND
5	Orange AF wire	RS485-A
6	Red AF wire	Standby
7	Black AF wire	Standby
19 core aviation plug		
No.	Color	Definition
1	Orange&white	Ethernet 1
2	Orange	
3	Green&white	
4	Green	
5	Blue&white	
6	Blue	
7	Brown&white	
8	Brown	
9	Orange&white	Ethernet 1
10	Orange	
11	Green&white	
12	Green	
13	Blue&white	
14	Blue	
15	Brown&white	
16	Brown	
17	Red 0.5AF wire	Standby
18	Black 0.5AF wire	Standby
19	-	-

USB3.0(left)		USB3.0(right)	
No.	Definition	No.	Definition
1	USB3.0 (1)	1	USB3.0 (2)

Top cables definition			
Left cable			
No.	Color	Definition	
1	RVSP4*0.3 Brown	DC12V/2A+	
2	RVSP4*0.3 Blue	DC12V/2A-	
3	RVSP4*0.3 Yellow	DC12V/2A+	
4	RVSP4*0.3 Black	DC12V/2A-	
Right cable			
No.	Color	Definition	
1	Orange&white	Ethernet 1	
2	Orange		
3	Green&white		
4	Green		
5	Blue&white		
6	Blue		
7	Brown&white		
8	Brown		
9	Orange&white	Ethernet 2	
10	Orange		
11	Green&white		
12	Green		
13	Blue&white		
14	Blue		
15	Brown&white		
16	Brown		
17	Brown 0.5AF wire	DC12V/2A+	
18	Blue 0.5AF wire	DC12V/2A-	
19	Red 0.5AF wire	DC12V/2A+	
20	Black 0.5AF wire	DC12V/2A-	
USB3.0(left)		USB3.0(right)	
No.	Definition	No.	Definition
1	USB3.0（1）	1	USB3.0（2）